
KTU
NOTES

AD FLOW STUDIES

ain inequality constraints(eg: load nodal ive power generation of the generators, s of a tap changing transformer etc)

al voltages and phase angles and hence ction at all the buses and power flows onnecting power channels.

esigning a new power system and for nsion of the existing one for increased *of the network equations.*

athematical technique for solution

ions.

node is associated with four

P, Q, V, δ

cified

uired to be obtained through the

he equations.

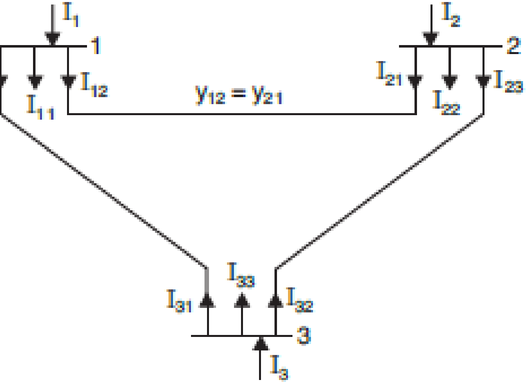
	Quantities specified	Quantities to be obtained
	P, Q	$\lvert V \rvert, \delta$
	$P, \lvert V \rvert$	Q, δ
	$\lvert V \rvert, \delta$	P, Q

...in unknown until the load flow ...omplete.

...ne of the generator buses is made ...dditional real and reactive power ...nsmission losses.

...ngle of the voltage at the slack bus ...ken as the reference.

System



$$+ I_{12} + I_{13}$$

$$+ (V_1 - V_2) y_{12} + (V_1 - V_3) y_{13}$$

$$+ y_{12} + y_{13}) - V_2 y_{12} - V_3 y_{13}$$

$$- V_2 Y_{12} + V_3 Y_{13}$$

$$+ y_{12} + y_{13}$$

$$y_{12}$$

$$y_{13}$$

$$\left.\right] = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & Y_{23} \\ Y_{31} & Y_{32} & Y_{33} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

nerally for 3 − bus system

$$= \sum_{q=1}^{3} Y_{pq} V_q , \quad p = 1 \text{ to } 3$$

nerally for n − bus system

$$= \sum_{q=1}^{n} Y_{pq} V_q , \quad p = 1,2,3,\ldots\ldots, n$$

trixform

$$\left.\right] = \begin{bmatrix} Y_{11} & : & Y_{1n} \\ : & : & Y_{2n} \\ Y_{n1} & : & Y_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ : \\ V_n \end{bmatrix}$$

tance matrix is a sparse matrix.

matrix along the leading diagonal.

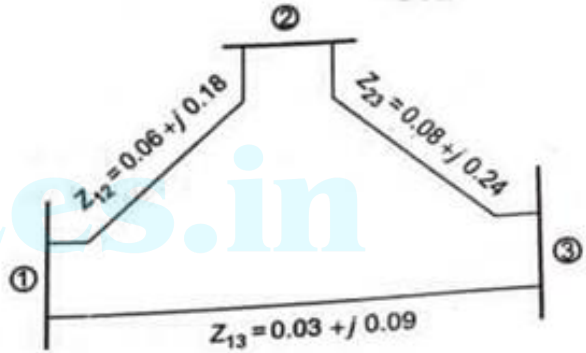
nemory requirements for storing
dmittance matrix is very low.

ment of each node is the sum of
nces connected to it. Off-diagonal
ne negated admittance between

Y_{bus} for the 3-bus system shown in Fig. . The line series
s :

Impedance (pu)
$0.06 + j\,0.18$
$0.03 + j\,0.09$
$0.08 + j\,0.24$

citances of the lines.



$0.06 + j\,0.18$

$= \dfrac{1}{0.03 + j\,0.09} = 3.33 - j\,10$

$= \dfrac{1}{0.08 + j\,0.24} = 1.25 - j\,3.75$

$+ y_{13} = 1.66 - j\,5 + 3.33 - j\,10 = 5 - j\,15$

$+ y_{23} = 1.66 - j\,5 + 1.25 - j\,3.75 = 2.91 - j\,8.75$

$+ y_{13} = 1.25 - j\,3.75 + 3.33 - j\,10 = 4.58 - j\,13.75$

$= -1.66 + j\,5 = Y_{21}$

$= -1.25 + j\,3.75 = Y_{32}$

$= -3.33 + j\,10 = Y_{31}$

$$\begin{matrix} 5 - j\,15 & -1.66 + j\,5 & -3.33 + j\,10 \\ 1.66 + j\,5 & 2.91 - j\,8.75 & -1.25 + j\,3.75 \\ 3.33 + j\,10 & -1.25 + j\,3.75 & 4.58 - j\,13.75 \end{matrix}\Bigg]$$

e diagonal element corresponding
s at which they are connected.

onal elements are unaffected

tances strengthen the diagonal
capacitances weaken it.

, each line has a total shunt admittance of $-j\,5.0$ pu. Determine
matrix.

etween buses 1 and 3, the total shunt admittance $= y_a + y_b = -j\,5.0$ and $y_a = y_b$
etween buses 2 and 3, the total shunt admittance $= y_e + y_f = -j\,5.0$ and $y_e = y_f$

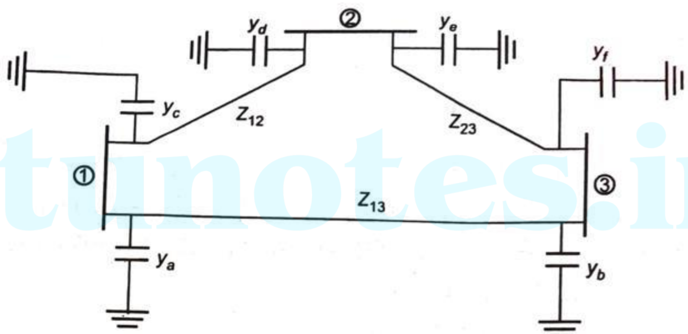


Fig. 14.4.

nsider the effect of shunt admittances, only the diagonal elements in Y_{bus} are
ve

$= y_c = y_d = y_e = y_f = -j\,2.5$

$= Y_{11\,(old)} + y_a + y_c = (5 - j\,15) + (-j\,5) = 5 - j\,20$

$= Y_{22\,(old)} + y_d + y_e = 2.91 - j\,8.75 - j\,5 = 2.91 - j\,13.5$

$= Y_{33\,(old)} + y_b + y_f = 4.58 - j\,13.75 - j\,5 = 4.58 - j\,18.75$

$$\begin{bmatrix} 5 - j\,20 & -1.66 + j\,5 & -3.33 + j\,10 \\ -1.66 + j\,5 & 2.91 - j\,13.5 & -1.25 + j\,3.75 \\ -3.33 + j\,10 & -1.25 + j\,3.75 & 4.58 - j\,18.75 \end{bmatrix}$$

$$Y_{pq}V_q, \quad p=1,2,3,\ldots\ldots,$$

$$+\sum_{\substack{q=1\\q\neq p}}^{n} Y_{pq}V_q$$

$$-\frac{1}{Y_{pp}}\sum_{\substack{q=1\\q\neq p}}^{n} Y_{pq}V_q \ldots\ldots\ldots\ldots(1)$$

$$P_P - jQ_p$$

$$\frac{jQ_p}{V_p^*}$$

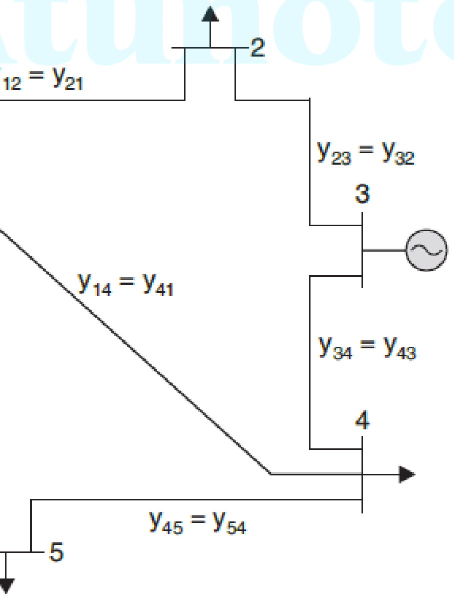
ng in Eqn(1)

$$\left[\frac{P_P - jQ_p}{V_p^*} - \sum_{\substack{q=1\\q\neq p}}^{n} Y_{pq}V_q\right], \quad p=1,2,3,\ldots\ldots,$$

ad flow equation with bus voltage as variable quations are nonlinear and they

d by an iterative method.

s system in the figure, Ybus is also



$$Y_{pq} = \begin{bmatrix} Y_{11} & Y_{12} & 0 & Y_{14} & Y_{15} \\ Y_{21} & Y_{22} & Y_{23} & 0 & 0 \\ 0 & Y_{32} & Y_{33} & Y_{34} & 0 \\ Y_{41} & 0 & Y_{43} & Y_{44} & Y_{45} \\ Y_{51} & 0 & 0 & Y_{54} & Y_{55} \end{bmatrix}$$

$$\left[\frac{P_2 - jQ_2}{V_2^*} - \left(Y_{21}V_1 + Y_{23}V_3\right)\right]$$

$$\left[\frac{P_3 - jQ_3}{V_3^*} - \left(Y_{32}V_2 + Y_{34}V_4\right)\right]$$

$$\left[\frac{P_4 - jQ_4}{V_4^*} - \left(Y_{41}V_1 + Y_{43}V_3 + Y_{45}V_5\right)\right]$$

$$\left[\frac{P_5 - jQ_5}{V_5^*} - \left(Y_{51}V_1 + Y_{54}V_4\right)\right]$$

table value of convergence criterion ε, olute value of the maximum change in en any two consecutive iterations is less cified tolerance ε, the convergence is he iterative procedure is terminated.

count K = 0

t p = 1

slack bus. If it is not a slack bus go to ce voltage at the slack bus is fixed both in nd phase, it does not vary during iterative d hence go to step 6 if it is a slack bus.

bus voltage V_{k+1}^{p} using equation and the bus voltage $\Delta V_p^K = V_p^{K+1} - V_p^K$.

nd ΔV_p^K.

uses have been taken into account. If
e next step, otherwise go back to

he largest absolute value of change in

ess than a specified tolerance ε,
flows and print the voltage and line
advance the iteration count K = K + 1
to step 3.

...of convergence in Gauss - Seidal ...ow as it requires a large numberof ... obtain the solution.

...e in this method can be speeded ...the acceleration factor.

...ng this factor, the voltage ...uring consecutive iteration is

$$^{+1)}_{cc)} = E_p^k + \propto \lfloor E_p^{(K+1)} - E_p^K \rfloor$$

alue of a for a system can be
running trial load flow studies.

recommended value of for most of
is 1.6.

oice of a may result in slower
e or even result in divergence from
.

feature

gnificant reduction in computer
ithmetic operations can be
n advance when they do not
the iterations

at a bus do not
terations

Let

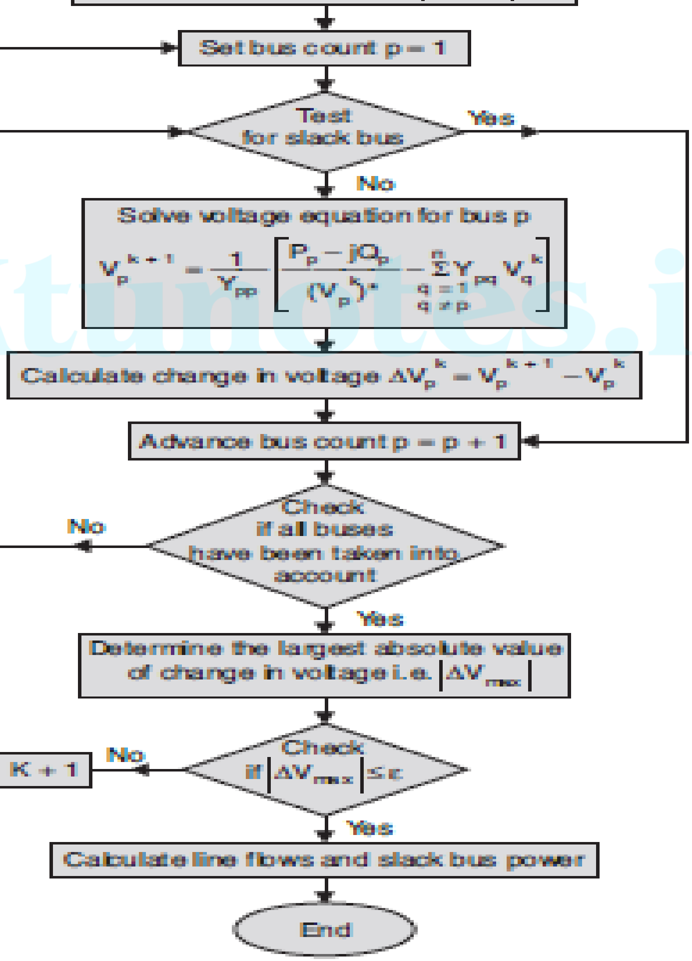
$$A_p = \frac{P_P - jQ_p}{Y_{pp}} \quad for\ all\ p = 1,2,....,n \quad \mathbf{p \neq s}$$

$r\ all\ \ p = 1,2,\ldots.,n \quad \mathbf{p \neq s}$

$q \neq p$

ation becomes

$$\Big)^{*} - \sum_{\substack{q=1 \\ q \neq p}}^{n} B_{pq} \mathrm{V_p}^{\mathrm{K}},\ p = 1,2,\ldots.,n \quad \mathbf{p \neq s}$$



Set bus count p = 1
Test for slack bus
Yes
No
Solve voltage equation for bus p
$V_p^{k+1} = \frac{1}{Y_{pp}}\left[\frac{P_p - jQ_p}{(V_p^k)^*} - \sum_{\substack{q=1\\q \neq p}}^{n} Y_{pq} V_q^k\right]$
Calculate change in voltage $\Delta V_p^k = V_p^{k+1} - V_p^k$
Advance bus count p = p + 1
Check if all buses have been taken into account
No
Yes
Determine the largest absolute value of change in voltage i.e. $|\Delta V_{max}|$
Check if $|\Delta V_{max}| \le \varepsilon$
No
K + 1
Yes
Calculate line flows and slack bus power
End

$$\frac{- jQ_2}{\left(V_2^k\right)^*} - \left(Y_{21}V_1 + Y_{23}V_3^k\right)\Bigg]$$

$$\frac{- jQ_3}{\left(V_3^k\right)^*} - \left(Y_{32}V_2^{k+1} + Y_{34}V_4^k\right)\Bigg]$$

$$\frac{- jQ_4}{\left(V_4^k\right)^*} - \left(Y_{41}V_1 + Y_{43}V_3^{k+1} + Y_{45}V_5^k\right)\Bigg]$$

$$\frac{- jQ_5}{\left(V_5^k\right)^*} - \left(Y_{51}V_1 + Y_{54}V_4^{k+1}\right)\Bigg]$$

tion

$$\frac{- jQ_p}{\left(V_p^k\right)^*} - \sum_{q=1}^{p-1} Y_{pq}V_q^{k+1} - \sum_{q=p+1}^{n} Y_{pq}V_q^k\Bigg]$$

esponding to a voltage controlled bus Q
calculated.

$$= \left\{ \sum_{q=1}^{n} f_p \left(e_q G_{pq} + f_q B_{pq} \right) - e_p \left(f_q G_{pq} - e_q B_{pq} \right) \right\}$$

$$ere\ V_p = e_p + jf_p \quad and \quad Y_{pq} = G_{pq} - jB_{pq}$$

rmissible (because of stability problem) or the maximum reactive power generation ecause of rotor heating).

he voltage controlled bus is made to act or that iteration only and the reactive ubstituted in the expression for that orrespond to the limit it has violated, *i.e., Qmin, the value to be substituted will be*

ed value of Q lies within the limits, this *be substituted in the expression.*

vo consecutive iterations is less than a
erance ε, the convergence is achieved and
ocedure is terminated.

ount K = 0

p = 1

lack bus. If it is a slack bus go to step
go to next step.

the buses are voltage controlled and which
For voltage controlled buses go to next
erwise go to step 4.

ue of the voltage magnitude of voltage
n that iteration by the specified value. Keep
e same as in that iteration.

orresponding to the voltage controlled bus.

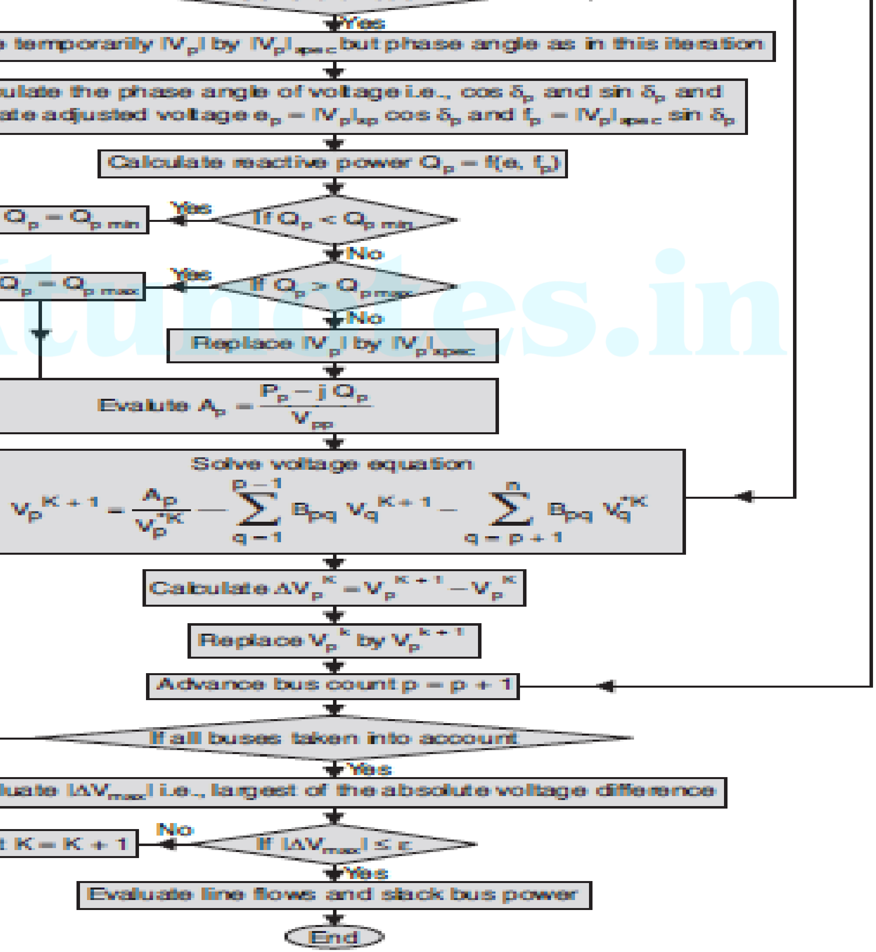
ew value of voltage for the bus. It is to be ere are more than one generator buses, the ude of that bus only is replaced by its while calculating *P and Q of a particular e of other generator buses will be o the* value in that iteration.

ll the voltage controlled buses violate the generation, the bus will be treated as a load gnitude of the reactive power at this bus to the limit it has violated, value of the oltage will correspond to the value in that orresponding to specified voltage),and then

s *have been taken into account.* If
ext step, otherwise go to step 1 (*c*).

e largest of the absolute value of
n voltage. If this is less than a
tolerance move on to next step,
o back to step 2.

e injected powers and the line
the nodal voltages.


Yes
temporarily |Vp| by |Vp|spec but phase angle as in this iteration
ulate the phase angle of voltage i.e., cos δp and sin δp and
ate adjusted voltage ep = |Vp|sp cos δp and fp = |Vp|spec sin δp
Calculate reactive power Qp = f(e, fp)
If Qp < Qp min
Yes
Qp = Qp min
No
If Qp > Qp max
Yes
Qp = Qp max
No
Replace |Vp| by |Vp|spec
Evalute Ap = (Pp − j Qp)/Vpp
Solve voltage equation
Calculate ΔVp^K = Vp^(K+1) − Vp^K
Replace Vp^k by Vp^(k+1)
Advance bus count p = p + 1
If all buses taken into account
Yes
luate |ΔVmax| i.e., largest of the absolute voltage difference
If |ΔVmax| ≤ ε
No
K = K + 1
Yes
Evaluate line flows and slack bus power
End

le and requires less memory space
it has some limitations too.

of load flow solution fails to
ystems having:

ber of radial transmission lines

aded transmission lines

alue of transfer admittances due to
of series capacitors in lines or three
sformers.

eatures, the Gauss - Seidalmethod
the load flow solution cannot be

ms without the above features, the
erations required for achieving
dependson the size of the system.
e system, greater will be the number

of slack bus also affects the number

–	–	1.06	Slack
0.5	0.2	$1 + j0.0$	*PQ*
0.4	0.3	$1 + j0.0$	*PQ*
0.3	0.1	$1 + j0.0$	*PQ*

e the voltages at the end of first iteration using Gauss-Seidel method. Take $\alpha = 1.6$.

: The admittance matrix will be as given below:

$$Y_{pq} = \begin{bmatrix} 3 - j12.0 & -2 + j8.0 & -1 + j4.0 & 0.0 \\ -2 + j8.0 & 3.666 - j14.664 & -0.666 + j2.664 & -1 + j4.0 \\ -1 + j4.0 & -0.666 + j2.664 & 3.666 - j14.664 & -2 + j8.0 \\ 0.0 & -1 + j4.0 & -2 + j8.0 & 3 - j12.0 \end{bmatrix}$$

rs for load buses are to be taken as negative and that for generator buses as

stem given

$$V_2{}^1 = \frac{1}{Y_{22}}\left[\frac{P_2 - jQ_2}{V_2^*} - Y_{21}V_1^0 - Y_{23}V_3^0 - Y_{24}V_4^0\right]$$

$$= \frac{1}{(3.666 - j14.664)}\left[\frac{-0.5 + j0.2}{1 - j0.0} - 106(-2 + j8) - 10\right.$$

$$\left.(-0.666 + j2.664) - (-1 + j4.0)10\right]$$

$$= (1.01187 - j0.02888)$$

$$V_{2\,\text{acc}}^1 = (1.0 + j0.0) + 1.6\{1.01187 - j0.02888 - 1.0 - j0.0\}$$

$$= 1.01899 - j0.046208 \quad \textbf{Ans.}$$

$$- (-0.666 + j2.664)(1.01899 - j0.046208) - (-2 + j8)(1 + j0.0)\Big]$$

$$9 - j0.029248$$

$$) + 1.6[0.994119 - j0.029248 - 1 - j0.0]$$

$$- j0.0467968 \quad \textbf{Ans.}$$

$$- jQ_4 \over V_4^{0*}} - Y_{42}V_2^1 - Y_{43}V_3^1\Big]$$

$$)\left[\frac{-0.3 + j0.1}{1 - j0.0} - (-1 + j4.0)(101899 - j0.046208)\right.$$

$$\left. - (-2 + j8)(0.99059 - j0.0467968)\right]$$

$$32 - j0.064684$$

$$0 + 1.6[0.9716032 - j0.064684 - 1 - j0.0]$$

$$5 - j0.1034944 \quad \textbf{Ans.}$$

$$\left(e_p + jf_p\right)^* \sum_{q=1}^{n} \left(G_{pq} - jB_{pq}\right)\left(e_q + jf_q\right)$$

$$\left(e_p - jf_p\right) \sum_{q=1}^{n} \left(G_{pq} - jB_{pq}\right)\left(e_q + jf_q\right)$$

real and imaginary parts

$$\left(e_q G_{pq} + f_q B_{pq}\right) + f_p \left(f_q G_{pq} - e_q B_{pq}\right)\Big\}$$

$$f_p \left(e_q G_{pq} + f_q B_{pq}\right) - e_p \left(f_q G_{pq} - e_q B_{pq}\right)\Big\}$$

$$f_p^2$$

load bus) and Pp and | Vp | for
s are specified and *ep and fp are*
antities.

system, the number of unknowns are

nson method is an iterative method
oximates the set of nonlinear
s equations to a set of linear
s equations using Taylor’s series

$$y_1 = f_1(x_1, x_2, \ldots\ldots x_n)$$
$$y_2 = f_2(x_1, x_2, \ldots\ldots x_n)$$

$$y_n = f_1(x_1, x_2, \ldots\ldots x_n)$$

equations we start with an approximate

$, x_n^0)$

are linearized about the initial guess.
uation *y1 = f1 and the result for the*
s will follow.

are the corrections required for $(x_1^0, x_2^0, \ldots, x_n^0)$
r the next better solution.

$$\delta x_1\big|_{x^0} \qquad \delta x_2\big|_{x^0} \qquad \delta x_n\big|_{x^0}$$

$$\begin{bmatrix} \ldots, x_n^0) \\ \ldots, x_n^0) \\ \\ \ldots, x_n^0) \end{bmatrix} = \begin{bmatrix} \frac{\delta f_1}{\delta x_1} & \frac{\delta f_1}{\delta x_2} & \cdots\cdots & \frac{\delta f_1}{\delta x_n} \\ \frac{\delta f_2}{\delta x_1} & \frac{\delta f_2}{\delta x_2} & \cdots\cdots & \frac{\delta f_2}{\delta x_n} \\ . \\ . \\ \frac{\delta f_n}{\delta x_1} & \frac{\delta f_n}{\delta x_2} & \cdots\cdots & \frac{\delta f_n}{\delta x_n} \end{bmatrix} \begin{bmatrix} \Delta x_1^0 \\ \Delta x_2^0 \\ . \\ . \\ \Delta x_n^0 \end{bmatrix}$$

of the equations requires

of left hand vector *B which is the*

f the specified quantities and

uantities at $(x_1^0, x_2^0, \ldots, x_n^0)$.

calculated at this guess.

he matrix equation gives $(\Delta x_1^0, \Delta x_2^0, \ldots, \Delta x_n^0)$

solution is obtained as

$$x_1^1 = x_1^0 + \Delta x_1^0$$
$$x_2^1 = x_2^0 + \Delta x_2^0$$
$$x_n^1 = x_n^0 + \Delta x_n^0$$

value or (*ii) the largest element in*
vector is less than a prespecified

$$
\begin{bmatrix}
\cdots\cdots & \frac{\delta P_n}{\delta e_n} & \frac{\delta P_n}{\delta f_2} & \frac{\delta P_n}{\delta f_3} & \cdots\cdots & \frac{\delta P_n}{\delta f_n} \\
\cdots\cdots & \frac{\delta Q_2}{\delta e_n} & \frac{\delta Q_2}{\delta f_2} & \frac{\delta Q_2}{\delta f_3} & \cdots\cdots & \frac{\delta Q_2}{\delta f_n} \\
\cdots\cdots & \frac{\delta Q_3}{\delta e_n} & \frac{\delta Q_3}{\delta f_2} & \frac{\delta Q_3}{\delta f_3} & \cdots\cdots & \frac{\delta Q_3}{\delta f_n} \\
\cdots\cdots & \frac{\delta Q_n}{\delta e_n} & \frac{\delta Q_n}{\delta f_2} & \frac{\delta Q_n}{\delta f_3} & \cdots\cdots & \frac{\delta Q_n}{\delta f_n}
\end{bmatrix}
\begin{bmatrix}
\Delta e_3 \\ \cdot \\ \cdot \\ \Delta e_n \\ \Delta f_2 \\ \Delta f_3 \\ \cdot \\ \cdot \\ \Delta f_n
\end{bmatrix}
$$

$-1)\times 2(n-1)$ $\qquad$ $2(n-1)\times 1$

$$\left.\begin{matrix} -- \\ | \; J_4 \end{matrix}\right] \begin{bmatrix} -- \\ \Delta f \end{bmatrix}$$

system contains all types of buses,

$$\left.\begin{matrix} J_1 \,|\, J_2 \\ --- \\ J_3 \,|\, J_4 \\ --- \\ J_5 \,|\, J_6 \end{matrix}\right] \begin{bmatrix} \Delta e \\ -- \\ \Delta f \end{bmatrix}$$

ne off - diagonal elements of J_2 are

$$\frac{\partial P_p}{\partial f_q} = e_p B_{pq} + f_p G_{pq}, \quad q \neq p$$

ne diagonal elements of J_2 are

$$\frac{\partial P_p}{\partial f_p} = 2 f_p G_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} f_p G_{pq} + e_q B_{pq}$$

ne off - diagonal elements of J_3 are

$$\frac{\partial Q_p}{\partial e_q} = e_p B_{pq} + f_p G_{pq}, \quad q \neq p$$

ne diagonal elements of J_3 are

$$\frac{\partial Q_p}{\partial e_p} = 2 e_p B_{pp} - \sum_{\substack{q=1 \\ q \neq p}}^{n} f_q G_{pq} - e_q B_{pq}$$

$$= 2 f_p B_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} e_q G_{pq} + f_q B_{pq}$$

off - diagonal elements of J_5 are

$$= 0, q \neq p$$

diagonal elements of J_5 are

$$= 2 e_p$$

off - diagonal elements of J_6 are

$$= 0, q \neq p$$

diagonal elements of J_5 are

$$= 2 f_p$$

, and | Vsp | be the specified
t the bus p.

suitable value of the solution (flat
ile in our case) the value of *P, Q*
the various buses are calculated.

$$\Delta P_p = P_{sp} - P_p^0$$

$$\Delta Q_p = Q_{sp} - Q_p^0$$

$$\Delta V_p \,|^2 = |\, V_{sp} \,|^2 - |\, V_p^0 \,|^2$$

lated the Jacobian matrix and the

umn vector corresponding tothe

(initial solution) the desired

oltage vector $\begin{bmatrix}\Delta e\\ \Delta f\end{bmatrix}$

lated by using any standard

tter solution will be

$$e_p^1 = e_p^0 + \Delta e_p^0$$

$$f_p^1 = f_p^0 + \Delta f_p^0$$

K = 0

bus. If yes, go to step 10.

nd reactive powers *Pp and Qp respectively*

question is a generator bus. If yes, compare the exceeds the limit, fix the reactive power Q_p^k orresponding limit and treat the bus as a load on and go to next step. *If the limit is not violated e residue.*

$|^2 = |V_p|_{spec}^2 - |V_p^k|^2$

he largest of the absolute value of the

of the absolute value of the residue is
 to step 17.

nents for Jacobian matrix.

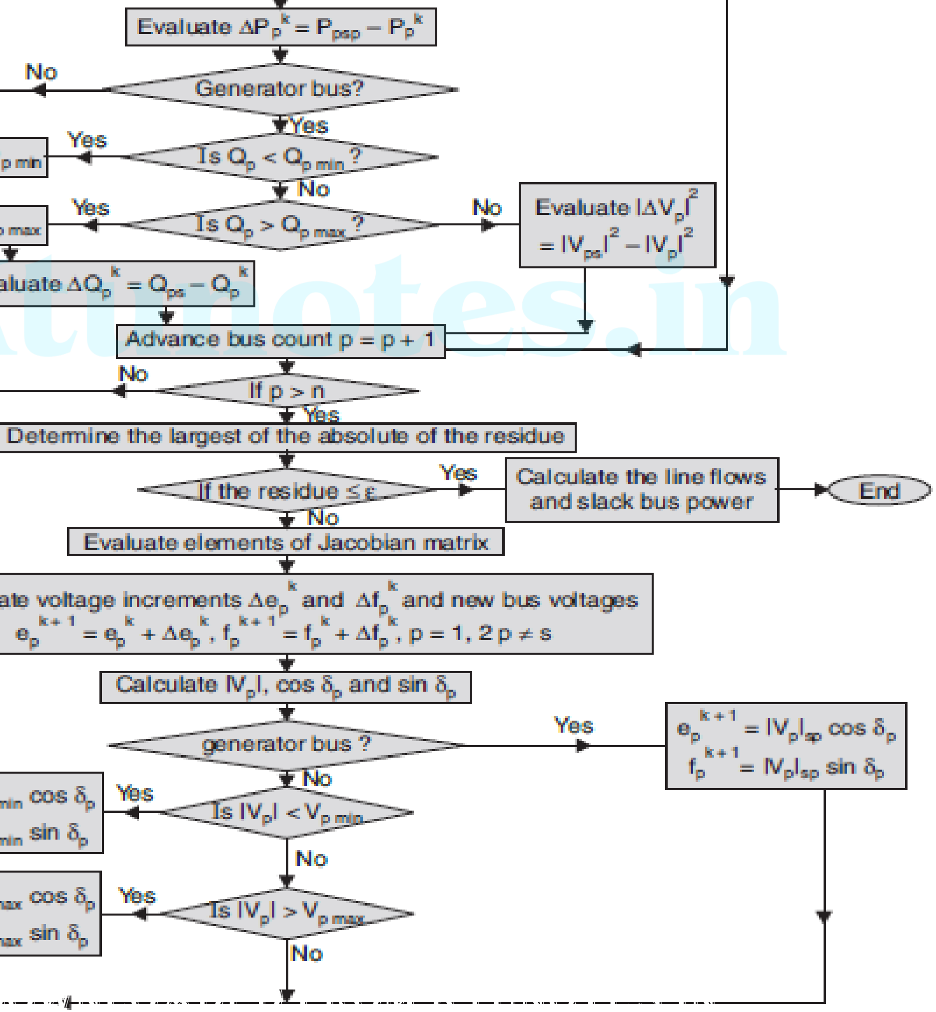
tage increments Δe_p^k and Δf_p^k.

v bus voltages $e_p^{k+1} = e_p^k + \Delta e_p^k$ and $f_p^{k+1} = f_p^k + \Delta f_p^k$.

nd sin δ of all voltages.

ation count *K = K + 1 and go to step 4.*

 and line powers and print the results.



Evaluate $\Delta P_p^k = P_{psp} - P_p^k$
Generator bus?
No
Yes
Is $Q_p < Q_{p\,min}$?
Yes
No
Is $Q_p > Q_{p\,max}$?
Yes
No
Evaluate $|\Delta V_p|^2 = |V_{ps}|^2 - |V_p|^2$
aluate $\Delta Q_p^k = Q_{ps} - Q_p^k$
Advance bus count p = p + 1
If p > n
No
Yes
Determine the largest of the absolute of the residue
If the residue ≤ ε
Yes
No
Calculate the line flows and slack bus power
End
Evaluate elements of Jacobian matrix
ate voltage increments Δe_p^k and Δf_p^k and new bus voltages
$e_p^{k+1} = e_p^k + \Delta e_p^k$, $f_p^{k+1} = f_p^k + \Delta f_p^k$, p = 1, 2 p ≠ s
Calculate $|V_p|$, $\cos \delta_p$ and $\sin \delta_p$
generator bus ?
Yes
No
$e_p^{k+1} = |V_p|_{sp} \cos \delta_p$
$f_p^{k+1} = |V_p|_{sp} \sin \delta_p$
Is $|V_p| < V_{p\,min}$
Yes
No
Is $|V_p| > V_{p\,max}$
Yes
No

voltages	MW	MVAR	MW	MVAR
$1.06 + j0.0$	0	0	0	0
$1.0 + j0.0$	0.2	0.0	0.0	0.0
$1.0 + j0.0$	0	0	0.6	0.25

$$y_{12} = \frac{1}{Z_{12}} = \frac{(0.08 - j0.24)}{(0.08 + j0.24)(0.08 - j0.24)}$$

$$= 1.25 - j3.75$$

$= 5 - j15$ and $y_{23} = 1.6667 - j5.0$ and the nodal admittance matrix

$$Y = \begin{bmatrix} 6.25 - j18.75 & -1.25 + j3.75 & -5 + j15 \\ -1.25 + j3.75 & 2.916 - j8.75 & -1.666 + j5.0 \\ -5 + j15 & -1.666 + j5.0 & 6.666 - j20 \end{bmatrix}$$

lat voltage profile for bus 2 and 3 and for bus 1.

$$V_1 = 1.06 + j0.0$$

lal admittance matrix and the assumed voltage solution,

$$B_{11} = 18.75 \qquad e_1 = 1.06 \qquad f_1 = 0.0$$

$$B_{12} = -3.75 \qquad e_2 = 1.0 \qquad f_2 = 0.0$$

$$B_{13} = -15.0 \qquad e_3 = 1.0 \qquad f_3 = 0.0$$

$$B_{22} = 8.75$$

$$B_{23} = -5.0$$

$$B_{33} = 20.0$$

$$P_2 = e_2(e_1G_{21} + f_1B_{21}) + f_2(f_1G_{21} - e_1B_{21}) + e_2(e_2G_{22} + f_2B_{22})$$
$$+ f_2(f_2G_{22} - e_2B_{22}) + e_2(e_3G_{23} + f_3B_{23}) + f_2(f_3G_{23} - e_3B_{23})$$

$$= 1.0\{1.06(-1.25) + 0.0\,(-3.75)\} + 0.0\{0.0\,(-1.25) - 1.06\,(-375)\}$$
$$+ 1.0\{1.0(2.916)\} + 0.0\{\ \} + 1.0\{1.0\,(-1.666)\} + 0.0\,\{\ \}$$

$$= -1.325 + 2.916 - 1.666 = -0.075$$

nents

$$\frac{\partial P_p}{\partial e_p} = 2e_p G_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} (e_q G_{pq} + f_q B_{pq})$$

$$\frac{\partial P_2}{\partial e_2} = 2e_2 G_{22} + e_1 G_{21} + f_1 B_{21} + e_3 G_{23} + f_3 B_{23}$$

$$= 2 \times 1.0 \times 2.916 + 1.06(-1.25) + 0.0(-3.75) + 1.0(-1.666) + 0.0(-5.0) = 2.848$$

$$\frac{\partial P_3}{\partial e_3} = 2e_3 G_{33} + e_1 G_{31} + f_1 B_{31} + e_2 G_{32} + f_2 B_{32}$$

$$= 2(1.0)(6.666) + 1.06(-5.0) + 0.0(-15.0) + 1.0\,(-1.666) + 0.0(-5.0) = 6.3666$$

$$\frac{\partial P_p}{\partial f_p} = 2f_p G_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} (f_q G_{pq} + e_q B_{pq})$$

$$\frac{\partial P_2}{\partial f_2} = 2f_2 G_{22} + f_1 G_{21} - e_1 B_{21} + f_3 G_{23} - e_3 B_{23}$$

$$= 2(0.0)(2.916) + (0.0)(G_{21}) - 1.06\,(-3.75) - 1.0(-5.0) = 8.975$$

$$\frac{\partial P_3}{\partial f_3} = 20.90$$

elements:

$$\frac{\partial P_p}{\partial e_q} = e_p G_{pq} - f_p B_{pq}$$

$$\frac{\partial P_2}{\partial e_3} = e_2 G_{23} - f_2 B_{23} = -1.666$$

$$\frac{\partial Q_p}{\partial e_p} = 2e_p B_{pp} - \sum_{\substack{q=1 \\ q \neq p}}^{n} (f_q G_{pq} - e_q B_{pq})$$

$$\frac{\partial Q_2}{\partial e_2} = 2e_2 B_{22} - f_1 G_{21} + e_1 B_{21} - f_3 G_{23} + e_3 B_{23}$$

$$= 2(1.0)(8.75) + 1.06\,(-3.75) + 1.0(-5.0) = 8.525$$

$$\frac{\partial Q_3}{\partial e_3} = 19.1$$

$$\frac{\partial Q_p}{\partial f_p} = 2f_p B_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} (e_q G_{pq} + f_q B_{pq})$$

$$\frac{\partial Q_2}{\partial f_2} = -2.991$$

$$\frac{\partial Q_3}{\partial f_3} = -6.966$$

al elements are calculated and the final set of linear equations at

$$\begin{bmatrix} 275 \\ 0.3 \\ 225 \\ .65 \end{bmatrix} = \begin{bmatrix} 2.846 & -1.666 & 8.975 & -5.0 \\ -1.666 & 6.366 & -5.0 & 20.90 \\ 8.525 & -5.0 & -2.991 & 1.666 \\ -5.0 & 19.1 & 1.666 & -6.966 \end{bmatrix} \begin{bmatrix} \Delta e_2 \\ \Delta e_3 \\ \Delta f_2 \\ \Delta f_3 \end{bmatrix}$$

se the reactive power constraint at bus 2 in the previous problem is
e the equations at the end of first iteration.

= − 0.225 and the lower limit is − 0.3, therefore, the bus 2 behaves

$$\Delta Q_3 = -0.25 - (-0.9) = 0.65$$

behaves like a generator bus therefore

$$|\Delta V_2|^2 = |V_{2\,sp}|^2 - |V_{2\,cal}|^2 = 1.04^2 - 1.0^2 = 0.0816$$

n elements corresponding to rows of ΔP_2, ΔP_3 and ΔQ_3 remain same as in those of Q_2 will change and they are calculated as follows:

$$\frac{\partial |V_2|^2}{\partial e_2} = 2e_2 = 2.0, \quad \frac{\partial |V_2|^2}{\partial e_3} = 0.0$$

$$\frac{\partial |V_2|^2}{\partial f_2} = 2f_2 = 0.0, \quad \frac{\partial |V_2|^2}{\partial f_3} = 0.0$$

quations will be

$$\begin{bmatrix} 0.275 \\ -0.3 \\ 0.65 \\ 0.0816 \end{bmatrix} = \begin{bmatrix} 2.846 & -1666 & 8.975 & -5.0 \\ -1666 & 6.366 & -5.0 & 20.90 \\ -5.0 & 19.1 & 1666 & -6.966 \\ 2.0 & 0.0 & 0.0 & 0.0 \end{bmatrix} \begin{bmatrix} \Delta e_2 \\ \Delta e_3 \\ \Delta f_2 \\ \Delta f_3 \end{bmatrix}$$

odal voltage affects the flow of reactive power
er practically does not change.

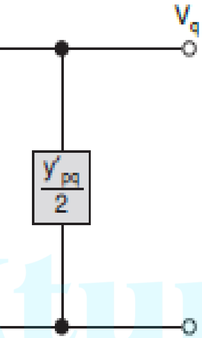
$$\left.\begin{matrix} 0 \\ {}_4 \end{matrix}\right]\begin{bmatrix} \Delta\delta \\ |\Delta V| \end{bmatrix}$$

to the elements of $\dfrac{\partial P}{\partial \delta}$ *which exists*

to the elements of $\dfrac{\partial P}{\partial |V|}$ *which do not exists*, $\therefore$ *zero.*

to the elements of $\dfrac{\partial Q}{\partial \delta}$ *hich do not exists*, $\therefore$ *zero*

to the elements of $\dfrac{\partial Q}{\partial |V|}$ *which exists.*



$$i_{pq} = (V_p - V_q)y_{pq} + V_p \frac{y'_{pq}}{2}$$

$$P_{pq} - jQ_{pq} = V^*_{pq}\ i_{pq}$$

$$P_{pq} - jQ_{pq} = V_p^* \left[(V_p - V_q)y_{pq} + V_p \frac{y'_{pq}}{2} \right]$$

$$= V_p^*(V_p - V_q)y_{pq} + V_p^* V_p \frac{y'_{pq}}{2}$$

he power flow from bus q to p is

$$_{qp} = V_q^*(V_q - V_p)y_{pq} + V_q^* V_q \frac{y'_{pq}}{2}$$

ne pq is the algebraic sum of the power flows $(P_{qp} - jQ_{pq})$ and

mulated in polar coordinates with
oximation which results into a fast
r load flow solution.

$q=1$

d $Admit \tan ce\ in\ polar\ coordinates\ represented\ as$

$$Y_{pq} = \left|Y_{pq}\right| e^{-j\theta pq}$$

g these values in equation

$$\left|V_p\right| e^{-j\delta p} \sum_{q=1}^{n} \left|Y_{pq}\right| e^{-j\theta pq} \left|V_q\right| e^{j\delta q}$$

$$_{pq}\big| e^{-j(\theta pq + \delta p - \delta q)}$$

$$V_q Y_{pq}\big| \cos\left(\theta pq + \delta p - \delta q\right)$$

$$V_q Y_{pq}\big| \sin\left(\theta pq + \delta p - \delta q\right)$$

$.n$

$$\sum_{\substack{q=1 \\ q\neq p}}$$

inearization can be rewritten in matrix form as

$$\left. |V| \right]$$

e elements of Jacobian matrix

decoupled load flow method is that real power changes s in voltage magnitude and are mainly sensitive to angle. Similarly, are less sensitive to change in angle but mainly sensitive to change these assumptions, reduces to

$$/|V|]$$

al element of H is

$$|V_p V_q Y_{pq}| \sin(\theta pq + \delta_p - \delta_q)$$

$$\cos(\delta_p - \delta_q) + G_{pq} \sin(\delta_p - \delta_q)]$$

- diagonal element of L is

$$= |V_p V_q Y_{pq}| \sin(\theta pq + \delta_p - \delta_q)$$

$$\cos(\delta_p - \delta_q) + G_{pq} \sin(\delta_p - \delta_q)]$$

equations, it is seen that $H_{pq} = L_{pq}$

$$V_q Y_{pq}\right|\sin\left(\theta pq+\delta_p-\delta_q\right)$$

nts of L are given as

$$^2 Y_{pp}\right|\sin\theta_{pp}+\sum_{\substack{q=1\\ q\neq p}}^{n}\left|V_p V_q Y_{pq}\right|\sin\left(\theta pq+\delta_p-\delta_q\right)$$

$$\cos(\delta_p - \delta_q) \simeq 1$$

$$G_{pq} \sin(\delta_p - \delta_q) \leq B_{pq}$$

$$Q_p << B_{pp} V_p^2$$

Jacobian elements now become

$$L_{pq} = H_{pq} = - | V_p V_q | B_{pq} \text{ for } q \neq p$$

$$L_{pp} = H_{pp} = - B_{pp} |V_p|^2$$

Computation time per iteration is less	Computation time per iteration is more	Computation time per iteration is less
It has linear convergence characteristics	It has quadratic convergence characteristics	....
The number of iterations required for convergence increases with size of the system	The number of iterations are independent of the size of the system	The number of iterations are does not dependent of the size of the system
Less memory requirements	More memory requirements.	Less memory requirements than N.R.method.